

Motor Sequence Experiment
PSY310 - Lab in Psychology
Lab Report

4th October 2025

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GitHub Link:

https://github.com/aryanm4/PSY310_LabInPsychology/tree/Motor-Sequence

Introduction

Motor sequence learning is a vital process in people's daily lives, as it makes them able to decide their actions with efficiency and accuracy based on predictions made from learned sequences. In the serial reaction time task, participants are expected to recognise a pattern in stimuli without conscious awareness of the pattern, where reaction time to respond decreases and accuracy increases when the stimuli are contingent.¹ These experiments investigate the implicit learning of sequences which takes place in the absence of direct instructions, whereas explicit learning requires active attention and detailed regulations.² Investigating this implicit learning method is more relevant to daily life, especially applying to the activities of growing children such as learning how to walk or ride a bicycle.³

¹ Forrin & MacLeod, 2017

² Dahms et al., 2019

³ Baars & Gage, 2010

Method

Participants

The conducted experiment had 5 participants. They were all undergraduates students at Ahmedabad University between the ages of 18 and 22. They were all male participants. They were fully informed of the experiment's objective, to click the correct key corresponding to the detected signal, before it was conducted and full consent was provided.

Participants were not informed about the firstly sequential and then random order of the trials before the experiment was conducted to avoid biases, however, they were completely debriefed after the experiment.

Materials and Procedure

This experiment was created on a desktop computer with a resolution of 1920x1200 pixels using PsychoPy-2025.1.1. The procedure recorded the participants' reaction time taken in seconds to click the key correctly corresponding to the position of the probe. 200 trials of the experiment were conducted per participant.

When the trial began, the screen presented four rectangles, size 100 pixels in height and 10 pixels in width, at positions (-150, 0), (-50, 0), (50, 0) and (150, 0) pixels. After one second, a triangular probe appears on top of one of the rectangles. The participants were instructed to click the 'z' key if the probe appeared on the first rectangle, the 'x' key if it appeared on the second, the 'c' key if it appeared on the third, and the 'v' key if it appeared on the fourth.

For the first 100 trials, the probe was made to appear on the four rectangles in a **sequential** order. It followed a set pattern, going from 'c' to 'z' to 'v' to 'x,' as shown in Figure 1. The participant was expected to recognise this sequence after two or three loops. In the next 100 trials, the probe was made to appear in a completely **random** sequence. In these trials, the reaction time of the participants was expected to increase.

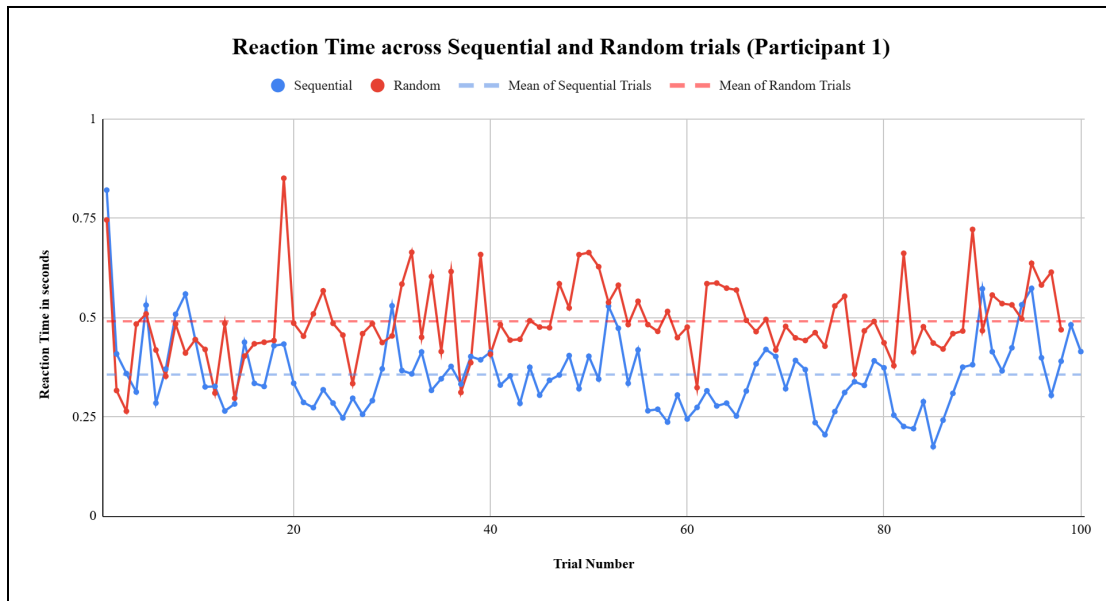
	A	B
1	pos	corr_resp
2	50	c
3	-150	z
4	150	v
5	-50	x

Figure 1 - Conditions and sequence created for the trials using an Excel format

Results

Participant 1

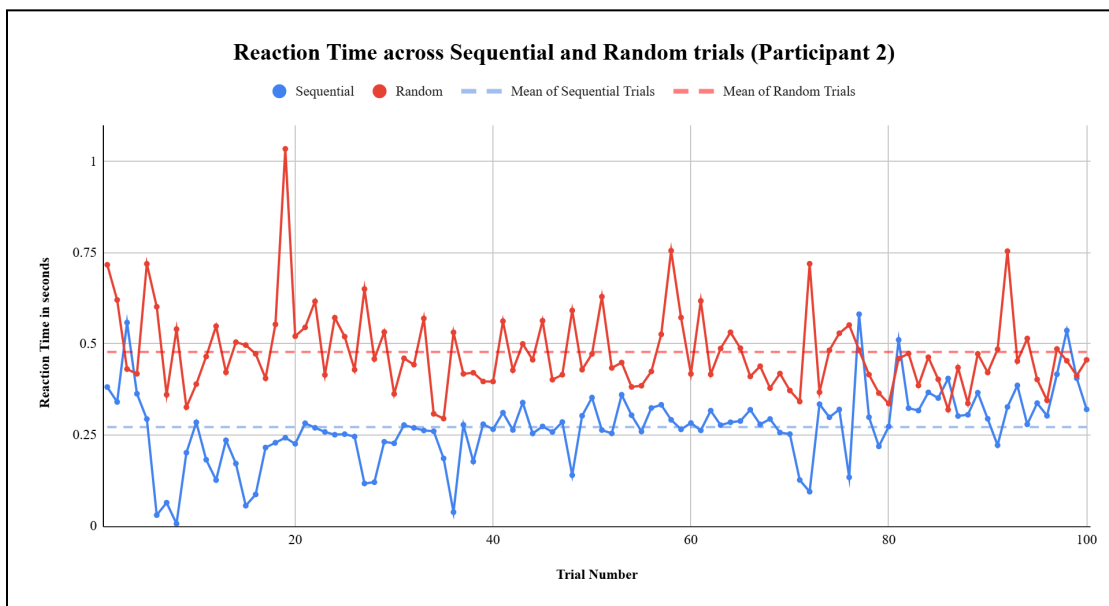
The mean reaction time of sequential trials was 0.3566 seconds, whereas for random trials it was 0.4905 seconds. This corresponds to a mean difference of 0.1339 seconds.



Graph of reaction time across sequential and random trials for participant 1

Participant 2

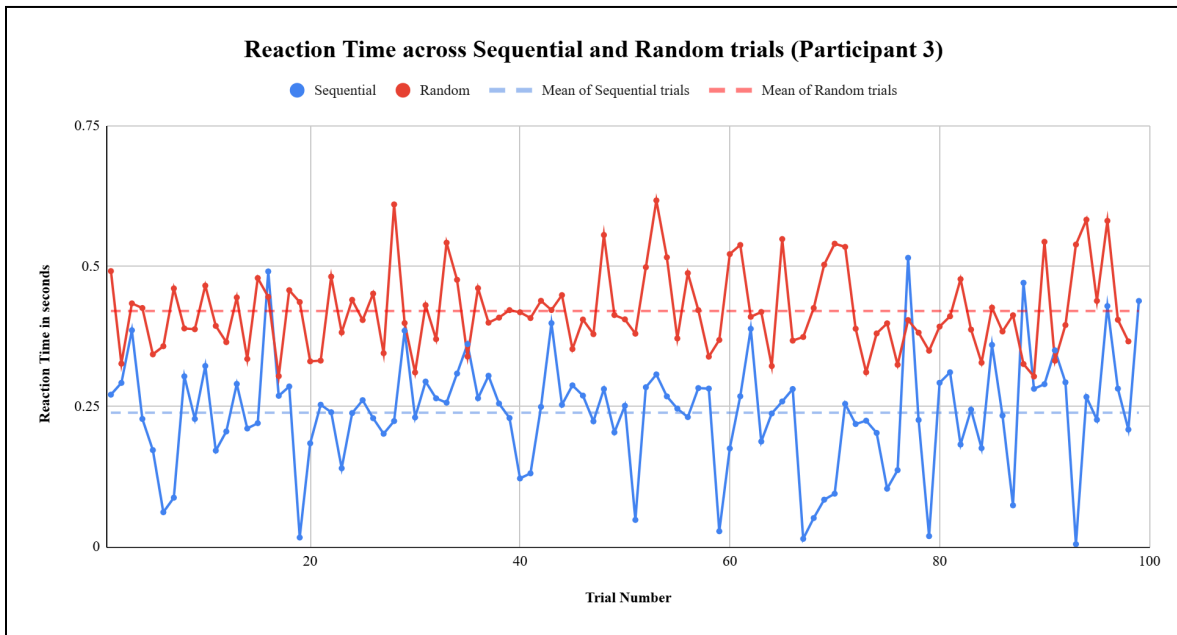
The mean reaction time for sequential trials was 0.2718 seconds, whereas for random trials it was 0.4779 seconds. This corresponds to a mean difference of 0.2061 seconds.



Graph of reaction time across sequential and random trials for participant 2

Participant 3

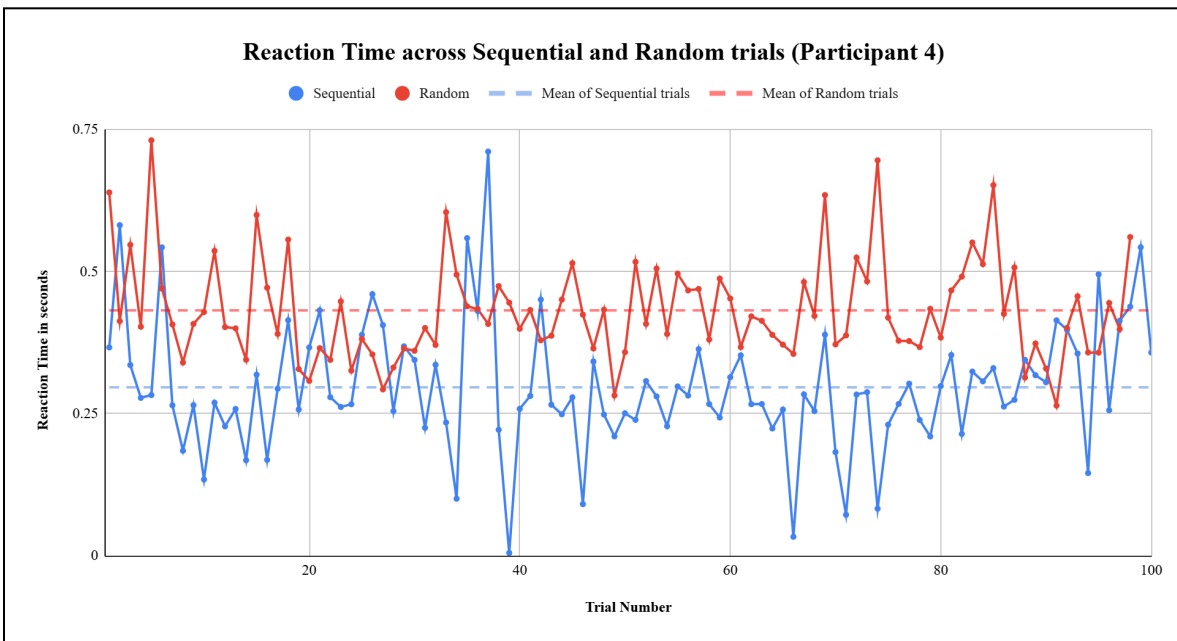
The mean reaction time of sequential trials was 0.2390 seconds, whereas for random trials it was 0.4204 seconds. This corresponds to a mean difference of 0.1814 seconds.



Graph of reaction time across sequential and random trials for participant 3

Participant 4

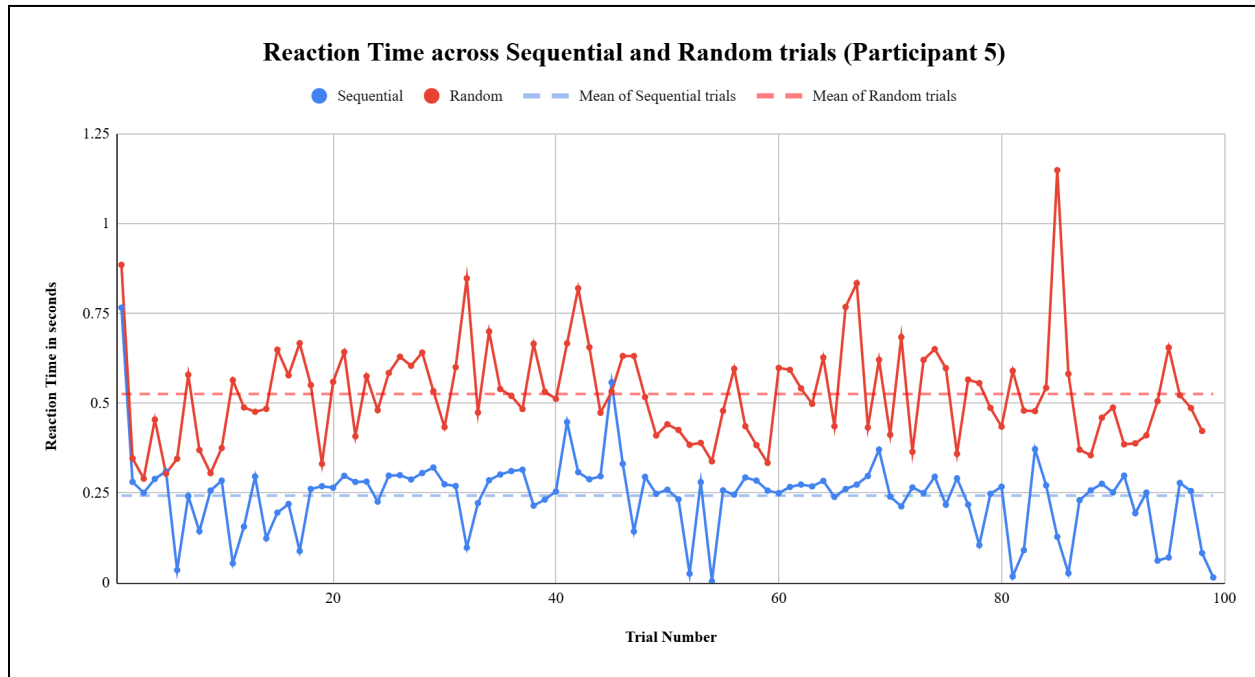
The mean reaction time of sequential trials was 0.2965 seconds, whereas for random trials it was 0.4204 seconds. This corresponds to a mean difference of 0.4317 seconds.



Graph of reaction time across sequential and random trials for participant 4

Participant 5

The mean reaction time of sequential trials was 0.2432 seconds, whereas for random trials it was 0.5259 seconds. This corresponds to a mean difference of 0.2827 seconds.



Graph of reaction time across sequential and random trials for participant 4

Across the 500 sequential and 500 random trials performed by all participants, the mean reaction time for sequential trials was calculated to be 0.2816 seconds, and for random trials it was 0.4693 seconds. This gives an overall mean difference of 0.1877 seconds.

Discussion

The results for each participant showed a significantly slower reaction time when the trials were randomly ordered compared to when they were sequential. This happens because in sequential trials, participants are trained to recognise the pattern and accordingly respond faster as it creates an expectation for the next stimulus. However, when this expectation is broken in randomly ordered trials, participants take longer to respond.

On the other hand, results from this study can not be generalised even though the applicability of this study to implicit learning in real life is high. Firstly, the ecological validity of this experiment is low, because clicking a key after detecting a stimulus can not be valid in real-life situations, but only in laboratory settings. Secondly, the design of this experiment is in a set order, where participants go through 100 sequential trials first and then 100 random trials. If this design is shifted to a random order, where trials from the random or sequential condition can be presented in any order, it will provide more insightful results as the increase or decrease in reaction time when the condition is changed would be observable. Therefore, further research is required.

References

Baars, B. J., & Gage, N. M. (2010). *Cognition, Brain, and Consciousness*. Elsevier.

Dahms, C., Brodoehl, S., Witte, O. W., & Klingner, C. M. (2019). The importance of different learning stages for motor sequence learning after stroke. *Human Brain Mapping*, 1, 270–286. <https://doi.org/10.1002/hbm.24793>

Forrin, N. D., & MacLeod, C. M. (2017). Contingency proportion systematically influences contingency learning. *Attention, Perception, & Psychophysics*, 155–165. <https://doi.org/https://doi.org/10.3758/s13414-017-1424-4>